

LO2 Activity: RPA Feasibility Case Study — Model Solution

Case Scenario

Global Logistics Corp operates **3 regional distribution hubs** and is experiencing high error rates in manual inventory reconciliation.

The company wants to determine which activities should be automated using **Robotic Process Automation (RPA)**.

TASK 1: RPA Feasibility Evaluation

Step 1: Four Candidate Tasks

Assume the company identifies these four processes:

Candidate Task	Current Process	Frequency	Main Problem
A. Inventory Reconciliation	Compare warehouse inventory with ERP records	Daily	Manual comparison and data-entry errors
B. Invoice Data Entry	Transfer supplier invoice information into ERP	Daily	Repetitive data entry
C. Exception Handling	Investigate inventory discrepancies	Daily	Requires human judgement
D. Inventory Report Distribution	Generate and email daily inventory reports	Daily	Repetitive reporting

Step 2: RPA Feasibility Matrix

Use a score from **1–5**:

- **1 = Very unsuitable**
- **5 = Highly suitable**

Criteria:

1. Rule-based
2. Transaction volume
3. Data standardisation
4. Error reduction potential
5. Implementation simplicity

For complexity, a **higher score means easier to implement**.

Criteria	A. Inventory Reconciliation	B. Invoice Data Entry	C. Exception Handling	D. Report Distribution
Rule-based	5	5	2	5
High transaction volume	5	4	3	5
Standardised data	4	4	2	5
Error reduction potential	5	5	4	3
Implementation simplicity	4	4	2	5
Total	23/25	22/25	13/25	23/25

Interpretation

A. Inventory Reconciliation — 23/25

Very suitable for RPA because:

- rules are clearly defined
- process occurs frequently
- data comes from structured systems
- manual errors are common
- the same process is repeated every day

B. Invoice Data Entry — 22/25

Also highly suitable because:

- invoices follow relatively standard formats
- employees repeatedly copy information
- data can be validated using predefined rules
- high volume creates significant labour savings

C. Exception Handling — 13/25

Less suitable for full RPA automation.

Why?

An inventory discrepancy might require someone to investigate:

"Why does the system show 100 units but the warehouse has only 92?"

Possible reasons include:

- damaged products
- missing products
- incorrect receiving records
- theft
- incorrect data entry
- returns

This requires **human judgement**.

RPA could identify and route the exception, but a human should investigate complex cases.

D. Inventory Report Distribution — 23/25

Highly suitable because:

- process is repetitive
- rules are clear
- reports use structured data
- reports are generated regularly
- emails can be automatically distributed

Recommended RPA Opportunities

For this case, the strongest candidates are:

1. Inventory Reconciliation

and

2. Inventory Report Distribution

Invoice Data Entry could also be implemented as another automation opportunity.

Note: The scores are an illustrative teaching model, not an actual assessment of a real company's processes.

TASK 2: ROI Calculation

Let's create realistic financial assumptions.

Current Manual Process

Global Logistics processes approximately:

10,000 inventory reconciliation transactions per month

Each manual reconciliation takes:

6 minutes

Therefore:

$10,000 \times 6 \text{ minutes} = 60,000 \text{ minutes}$

$60,000 \text{ minutes} \div 60 = 1,000 \text{ staff hours/month}$

Assume the average labour cost is:

\$24 per hour

Therefore:

$1,000 \times \$24 = \$24,000/\text{month}$

Current processing cost

\$24,000 per month

or

\$288,000 per year

After RPA Implementation

Suppose the RPA bot can process a transaction in:

1 minute

$10,000 \times 1 \text{ minute}$:

= 10,000 minutes

= 166.7 hours

$166.7 \times \$24$

= \$4,000/month

Therefore:

Manual cost

\$24,000/month

Automated cost

\$4,000/month

Monthly saving

$\$24,000 - \$4,000 = \$20,000$

Annual saving

$\$20,000 \times 12 = \$240,000$

Cost per Transaction

Before RPA

$\$24,000 \div 10,000$

= \$2.40 per transaction

After RPA

$\$4,000 \div 10,000$

= \$0.40 per transaction

Cost reduction

$\$2.40 - \0.40

= \$2.00 per transaction

Percentage reduction:

$$\frac{2.40 - 0.40}{2.40} \times 100$$

= 83.3% reduction

So the students can conclude:

RPA reduces the estimated processing cost from \$2.40 to \$0.40 per transaction, representing an approximately 83% reduction in labour processing cost.

ROI Calculation

Assume the company spends:

- RPA software and implementation: **\$80,000**
- Training and testing: **\$20,000**

Total first-year investment:

\$100,000

Annual saving:

\$240,000

Simple first-year ROI:

$$ROI = \frac{Savings - Investment}{Investment} \times 100$$
$$ROI = \frac{240,000 - 100,000}{100,000} \times 100$$

ROI = 140%

Therefore:

The illustrative first-year ROI is 140%.

The simple payback period is:

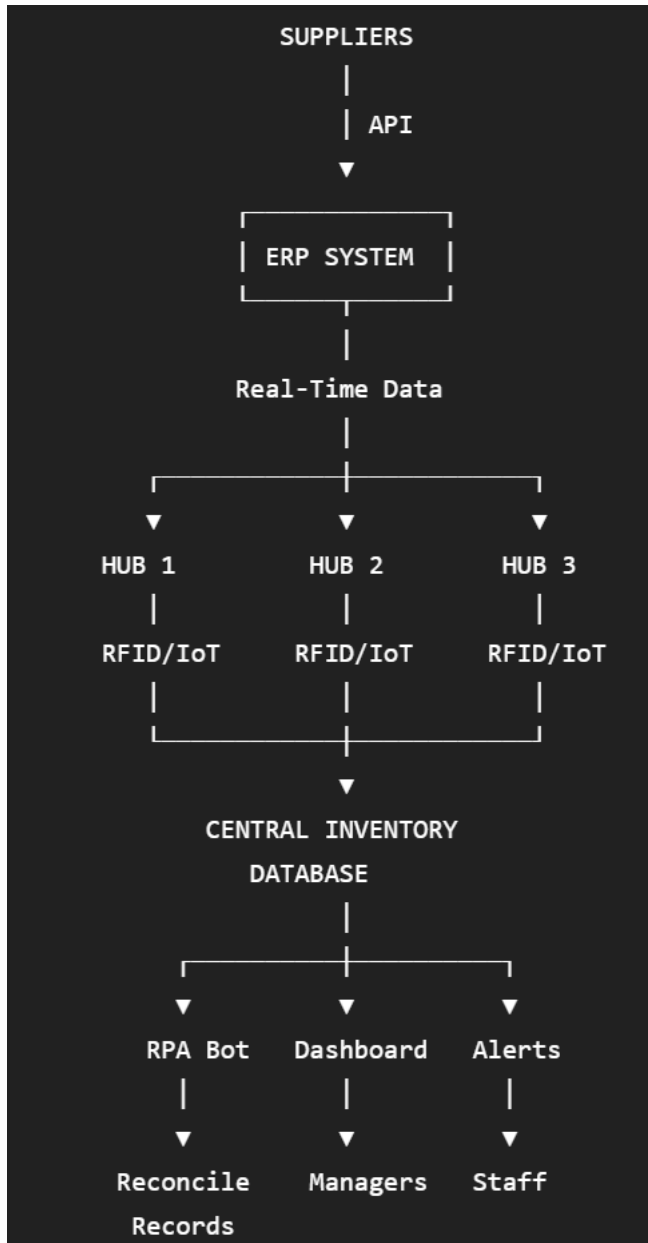
$$100,000 \div 20,000 = 5 \text{ months}$$

Payback period ≈ 5 months

TASK 3: Real-Time Inventory Tracking Integration Model

The company operates three regional distribution hubs.

The recommended architecture is:



How the System Works

Step 1 — Product arrives

Products arrive at one of the three distribution hubs.

Example:

Hub 1 receives 500 wireless keyboards.

RFID scanners automatically identify the products.

Step 2 — Inventory is updated

The RFID system sends information to the inventory system.

Instead of:

Employee manually enters 500 units

the system automatically records:

Wireless Keyboard: +500

Step 3 — Real-Time Monitoring

Managers can see:

Hub	Product	Stock
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Hub 1	Wireless Keyboard	500
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Hub 2	Wireless Keyboard	280
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Hub 3	Wireless Keyboard	150
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The company therefore has a central view of inventory.

Step 4 — RPA Reconciliation

The RPA bot compares:

Physical/RFID Inventory

against

ERP Inventory

Example:

Product	RFID Count	ERP Count	Difference
Keyboard	500	500	0
Mouse	300	295	5
Monitor	120	120	0
Laptop	80	75	5

The RPA bot automatically identifies:

 **Mouse discrepancy: 5 units**

 **Laptop discrepancy: 5 units**

Step 5 — Exception Handling

This is where humans become important.

The RPA bot should not automatically decide why the discrepancy occurred.

Instead:

RPA detects discrepancy

↓

Creates exception ticket

↓

Warehouse staff investigates

↓

Possible reasons:

- **damaged goods**
- **missing stock**
- **incorrect receiving**
- **incorrect shipment**
- **returned goods not recorded**
- **data-entry error**

↓

Staff corrects the record

↓

RPA updates reconciliation status

Step 6 — Automatic Alerts

The system can also establish rules.

For example:

If inventory falls below 50 units → Send alert.

If stock reaches:

45 units

the system automatically sends:

⚠ LOW INVENTORY: Wireless Keyboard – Hub 1 – 45 units remaining.

The company can then initiate replenishment.